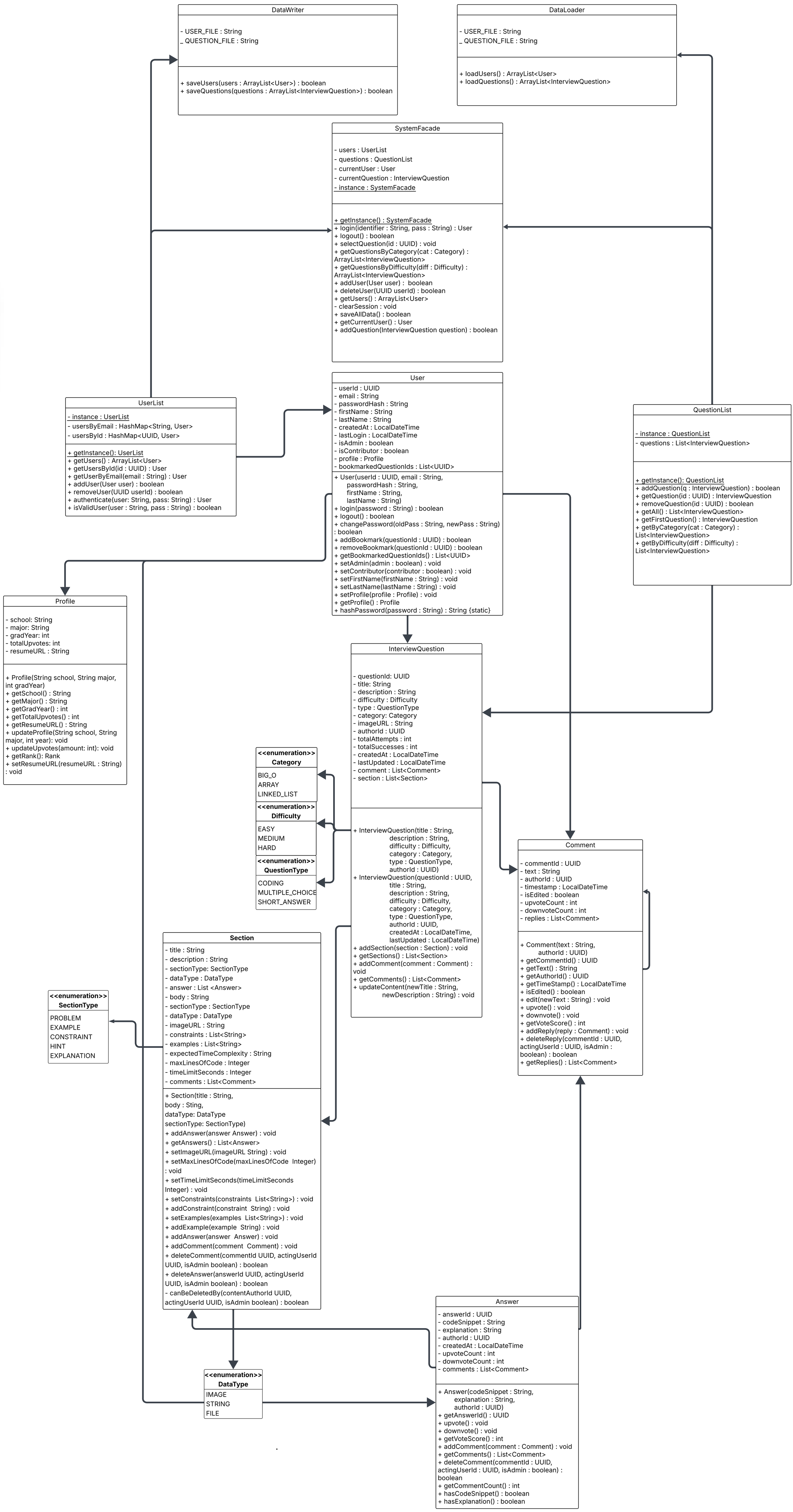
DataWriter
- USER_FILE : String
_ QUESTION_FILE : String
+ saveUsers(users : ArrayList<User>) : boolean
+ saveQuestions(questions : ArrayList<InterviewQuestion>) : boolean
DataLoader
- USER_FILE : String
_ QUESTION_FILE : String
+ loadUsers() : ArrayList<User>
+ loadQuestions() : ArrayList<InterviewQuestion>
SystemFacade
- users : UserList
- questions : QuestionList
- currentUser : User
- currentQuestion : InterviewQuestion
- instance : SystemFacade
+ getInstance() : SystemFacade
+ login(identifier : String, pass : String) : User
+ logout() : boolean
+ selectQuestion(id : UUID) : void
+ getQuestionsByCategory(cat : Category) : ArrayList<InterviewQuestion>
+ getQuestionsByDifficulty(diff : Difficulty) : ArrayList<InterviewQuestion>
+ addUser(User user) : boolean
+ deleteUser(UUID userId) : boolean
+ getUsers() : ArrayList<User>
- clearSession : void
+ saveAllData() : boolean
+ getCurrentUser() : User
+ addQuestion(InterviewQuestion question) : boolean
UserList
- instance : UserList
- usersByEmail : HashMap<String, User>
- usersById : HashMap<UUID, User>
+ getInstance(): UserList
+ getUsers() : ArrayList<User>
+ getUsersById(id : UUID) : User
+ getUserByEmail(email : String) : User
+ addUser(User user) : boolean
+ removeUser(UUID userId) : boolean
+ authenticate(user: String, pass: String) : User
+ isValidUser(user : String, pass : String) : boolean
User
- userId : UUID
- email : String
- passwordHash : String
- firstName : String
- lastName : String
- createdAt : LocalDateTime
- lastLogin : LocalDateTime
- isAdmin : boolean
- isContributor : boolean
- profile : Profile
- bookmarkedQuestionIds : List<UUID>
+ User(userId : UUID, email : String, passwordHash : String, firstName : String, lastName : String)
+ login(password : String) : boolean
+ logout() : boolean
+ changePassword(oldPass : String, newPass : String) : boolean
+ addBookmark(questionId : UUID) : boolean
+ removeBookmark(questionId : UUID) : boolean
+ getBookmarkedQuestionIds() : List<UUID>
+ setAdmin(admin : boolean) : void
+ setContributor(contributor : boolean) : void
+ setFirstName(firstName : String) : void
+ setLastName(lastName : String) : void
+ setProfile(profile : Profile) : void
+ getProfile() : Profile
+ hashPassword(password : String) : String {static}
QuestionList
- instance : QuestionList
- questions : List<InterviewQuestion>
+ getInstance(): QuestionList
+ addQuestion(q : InterviewQuestion) : boolean
+ getQuestion(id : UUID) : InterviewQuestion
+ removeQuestion(id : UUID) : boolean
+ getAll() : List<InterviewQuestion>
+ getFirstQuestion() : InterviewQuestion
+ getByCategory(cat : Category) : List<InterviewQuestion>
+ getByDifficulty(diff : Difficulty) : List<InterviewQuestion>
Profile
- school: String
- major: String
- gradYear: int
- totalUpvotes: int
- resumeURL : String
+ Profile(String school, String major, int gradYear)
+ getSchool() : String
+ getMajor() : String
+ getGradYear() : int
+ getTotalUpvotes() : int
+ getResumeURL() : String
+ updateProfile(String school, String major, int year): void
+ updateUpvotes(amount: int): void
+ getRank(): Rank
+ setResumeURL(resumeURL : String) : void
InterviewQuestion
- questionId: UUID
- title: String
- description : String
- difficulty : Difficulty
- type : QuestionType
- category: Category
- imageURL : String
- authorId : UUID
- totalAttempts : int
- totalSuccesses : int
- createdAt : LocalDateTime
- lastUpdated : LocalDateTime
- comment : List<Comment>
- section : List<Section>
+ InterviewQuestion(title : String, description : String, difficulty : Difficulty, category : Category, type : QuestionType, authorId : UUID)
+ InterviewQuestion(questionId : UUID, title : String, description : String, difficulty : Difficulty, category : Category, type : QuestionType, authorId : UUID, createdAt : LocalDateTime, lastUpdated : LocalDateTime)
+ addSection(section : Section) : void
+ getSections() : List<Section>
+ addComment(comment : Comment) : void
+ getComments() : List<Comment>
+ updateContent(newTitle : String, newDescription : String) : void
<<enumeration>> Category
BIG_O
ARRAY
LINKED_LIST
<<enumeration>> Difficulty
EASY
MEDIUM
HARD
<<enumeration>> QuestionType
CODING
MULTIPLE_CHOICE
SHORT_ANSWER
Comment
- commentId : UUID
- text : String
- authorId : UUID
- timestamp : LocalDateTime
- isEdited : boolean
- upvoteCount : int
- downvoteCount : int
- replies : List<Comment>
+ Comment(text : String, authorId : UUID)
+ getCommentId() : UUID
+ getText() : String
+ getAuthorId() : UUID
+ getTimeStamp() : LocalDateTime
+ isEdited() : boolean
+ edit(newText : String) : void
+ upvote() : void
+ downvote() : void
+ getVoteScore() : int
+ addReply(reply : Comment) : void
+ deleteReply(commentId : UUID, actingUserId : UUID, isAdmin : boolean) : boolean
+ getReplies() : List<Comment>
<<enumeration>> SectionType
PROBLEM
EXAMPLE
CONSTRAINT
HINT
EXPLANATION
Section
- title : String
- description : String
- sectionType: SectionType
- dataType : DataType
- answer : List <Answer>
- body : String
- sectionType : SectionType
- dataType : DataType
- imageURL : String
- constraints : List<String>
- examples : List<String>
- expectedTimeComplexity : String
- maxLinesOfCode : Integer
- timeLimitSeconds : Integer
- comments : List<Comment>
+ Section(title : String, body : Sting, dataType: DataType sectionType: SectionType)
+ addAnswer(answer Answer) : void
+ getAnswers() : List<Answer>
+ setImageURL(imageURL String) : void
+ setMaxLinesOfCode(maxLinesOfCode Integer) : void
+ setTimeLimitSeconds(timeLimitSeconds Integer) : void
+ setConstraints(constraints List<String>) : void
+ addConstraint(constraint String) : void
+ setExamples(examples List<String>) : void
+ addExample(example String) : void
+ addAnswer(answer Answer) : void
+ addComment(comment Comment) : void
+ deleteComment(commentId UUID, actingUserId UUID, isAdmin boolean) : boolean
+ deleteAnswer(answerId UUID, actingUserId UUID, isAdmin boolean) : boolean
- canBeDeletedBy(contentAuthorId UUID, actingUserId UUID, isAdmin boolean) : boolean
<<enumeration>> DataType
IMAGE
STRING
FILE
Answer
- answerId : UUID
- codeSnippet : String
- explanation : String
- authorId : UUID
- createdAt : LocalDateTime
- upvoteCount : int
- downvoteCount : int
- comments : List<Comment>
+ Answer(codeSnippet : String, explanation : String, authorId : UUID)
+ getAnswerId() : UUID
+ upvote() : void
+ downvote() : void
+ getVoteScore() : int
+ addComment(comment : Comment) : void
+ getComments() : List<Comment>
+ deleteComment(commentId : UUID, actingUserId : UUID, isAdmin : boolean) : boolean
+ getCommentCount() : int
+ hasCodeSnippet() : boolean
+ hasExplanation() : boolean